

Pattern 0 Oblivious Culture

Other names: This pattern doesn't exist in Crosby's, Humphrey's, or Curtis's models.

View of themselves: "We don't even know that we're performing a process."

Metaphor: Walking: When we want to go somewhere, we just stand up and go.

Management understanding and attitude: There is no comprehension that quality is a management issue.

Problem handling: Problems are suffered in silence.

Summation of quality position: "We don't have quality problems."

When this pattern is successful: To succeed, individuals need three conditions or beliefs:

- ✓ "I'm solving my own problems."
- ✓ "Those problems aren't too big for what I know is technically possible."
- ✓ "I know what I want better than anyone else."

Process results: Results depend totally on the individual. No records are kept, so there are no measurements. Because the customer is the developer, delivery is always acceptable.

Pattern 1 Variable Culture

Other names:

Crosby: Uncertainty Stage
Humphrey: Initial Process
Curtis: Herded

View of themselves: "We do whatever we feel like at the moment."

Metaphor: Riding a horse: When we want to go somewhere, we saddle up and ride...if the horse cooperates.

Management understanding and attitude: There is no comprehension of quality as a management tool.

Problem handling: Problems are fought with inadequate definition and no resolution (but with lots of yelling and accusations).

Summation of quality position: "We don't know why we have quality problems."

When this pattern is successful: To succeed, the individuals (or teams) need three conditions or beliefs:

- ✓ "I have great rapport with my customer."
- ✓ "I'm a competent professional."
- ✓ "My customer's problem isn't too big for me."

Process results: The work is generally one-on-one between customer and developer. Quality is measured internally by its function ("It works!"), externally by the working relationship. Emotion, personal relations, and mysticism drive everything. There is no consistent design, randomly structured code, and errors removed by haphazard testing. Some of the work is excellent, some is bizarre, and it all depends on the individual.

Pattern 2 Routine Culture

Other names:

Crosby: Awakening Stage
Humphrey: Repeatable Process
Curtis: Managed

View of themselves: "We follow our routines (except when we lose our nerve)."

Metaphor: A train: When we want to go somewhere, we find a train, which has large capacity and is very efficient...if we go where the tracks are. We're helpless when off the tracks.

Management understanding and attitude: There is a recognition that quality management may be of value, but there is no willingness to provide money or time to make it all happen.

Problem handling: Teams are set up to handle major problems. Long-range solutions are not solicited.

Summation of quality position: "Is it absolutely necessary to have problems with quality? Maybe if we just don't deal with them, the problems will go away."

When this pattern is successful: To succeed, people in these organizations need four conditions or beliefs:

- ✓ "We realize the problem is bigger than one small team can handle."
- ✓ "The problem is not too big for us to handle."
- ✓ "The developers must conform to our Routine process."
- ✓ "We hope we don't run into anything too exceptional."

Process results: The Routine organization has procedures to coordinate efforts, though its members only go through the motions of following them. Statistics on past performance are used not to change, but to prove that they are doing everything in the only reasonable way. Quality is measured internally by the numbers of errors ("bugs"). Generally, the organization uses bottom-up design and semi-structured code, with errors removed by testing and fixing. Routine organizations have many successes, but a few very large failures.

Pattern 3 Steering Culture

Other names:

Crosby: Enlightenment Stage
Humphrey: Defined Process
Curtis: Tailored

View of themselves: "We choose among our routines based on the results they produce."

Metaphor: A van: We have a large choice of destinations, but we must generally stay on mapped roads, and must be steered to stay on the road.

Management understanding and attitude: There is comprehension of quality as a management tool: "Through our quality program, we learn more about quality management, and become more supportive and helpful."

Problem handling: Problems are faced openly and resolved in an orderly way.

Summation of quality position: "Through commitment and quality improvement, we are identifying and resolving our problems."

When this pattern is successful: To succeed, people in these organizations need four conditions or beliefs:

- ✓ "The problem is big enough that we know a simple routine won't work."
- ✓ "Our managers can negotiate with the external environment."
- ✓ "We don't accept arbitrary schedules and constraints."
- ✓ "We are challenged, but not excessively."

Process results: They have procedures that are always well understood, but not always well defined in writing, and that are followed even in crisis. Quality is measured by user (customer) response, but not systematically. Some measuring is done, but everybody debates which measurements are meaningful. Typically they use top-down design, structured code, design and code inspections, and incremental releases. The organization has consistent success when it commits to undertake something.

Pattern 4 Anticipating Culture

Other names:

Crosby: Wisdom Stage
Humphrey: Managed Process
Curtis: Institutionalized

View of themselves: "We establish routines based on our past experience with them."

Metaphor: An airplane: When going somewhere, we can travel fast, reliably, and anywhere there's a field, but going this way requires a large initial investment.

Management understanding and attitude: There is understanding of the absolutes of quality management, and recognition of their personal role in this continuing emphasis.

Problem handling: Problems are identified early in their development. All functions are open to suggestion and improvement.

Summation of quality position: "Defect prevention is a routine part of our operation."

When this pattern is successful: To succeed, these organizations need three conditions or beliefs:

- ✓ "We have procedures, which we follow and try to improve."
- ✓ "We measure quality and cost (internally) by meaningful statistics."
- ✓ "We have an explicit process group to aid in the process."

Process results: They use sophisticated tools and techniques, including functional, theoretical design, mathematical verification, and reliability measurement. They have consistent success even on ambitious projects.

Pattern 5 Congruent Culture

Other names:

Crosby: Certainty Stage
Humphrey: Optimizing Process
Curtis: Optimized

View of themselves: "Everyone is involved in improving everything all the time."

Metaphor: The Starship Enterprise: When going somewhere, we can go where no one has gone before, we can carry anything, and we can beam ourselves anywhere, but this is all science fiction.

Management understanding and attitude: Quality management is considered an essential part of the company system.

Problem handling: Except in the most unusual cases, problems are prevented.

Summation of quality position: "We know why we do not have quality problems."

When this pattern is successful: To succeed, these organizations need three conditions or beliefs:

- ✓ "We have procedures, which we improve continuously."
- ✓ "We identify and measure all key process variables automatically."
- ✓ "Our goal is customer satisfaction, which drives everything."

Process results: Here are all of the good things achievable by the other patterns, plus the willingness to spend to reach the next level of quality. Quality is measured by customer satisfaction and by the mean time to customer failure (ten to one hundred years). Customers love the quality, and can bet their life on it. In some sense, Pattern 5 is like Pattern 0 in being totally responsive to the customer, but it is much better at what it does.